EASYPROPERTY

DETAILED PROJECT PROPOSAL

Real-Estate Portal | Full-Stack Web Application

Project Name	EasyProperty – Real-Estate Portal
Student Name	Huzaifa
Roll No	24-CIT-135
Starting Date	9 October 2026
End Date	9 March 2027
Duration	5-Months

1. Project Description

EasyProperty is a web-based real-estate portal designed for Pakistan. It will help users explore, compare, advertise, and manage residential properties, rental homes, commercial/business spaces, agricultural land, and investment-focused properties. It will provide separate dashboards and workflows for buyers, sellers, agents, landlords, and administrators. Planned features include property search and filters, favorites, enquiries, admin listing review, AI-assisted risk warnings, an AI legal information assistant, and 360-degree property tours. Payment functionality will initially be demonstrated through a sandbox or prototype unless an approved payment provider is available.

Users can browse properties, search and filter listings, view details, save favorites, compare properties, send enquiries, and request viewings. Sellers and agents can manage listings; landlords can organize rental records; investors can compare available listing information; and admins can review listings and reports.

2. Tools / Tech Stack

Technology / Tool	Purpose
HTML5	Semantic structure for pages, navigation, forms, and accessible content.
CSS3	Responsive layouts, styling, typography, spacing, and mobile/tablet support.
JavaScript ES6+	Frontend interactions, validation, and API requests.
React.js	Selected frontend framework for reusable components, routes, pages, and dashboards.
Vite	React development server and production build tool.
Node.js	JavaScript runtime for server-side development.
Express.js	REST API routes, middleware, validation, and error handling.
MongoDB	Selected database for users, properties, favorites, enquiries, viewing requests, reports, and status records.
Mongoose	Schemas, data validation, indexes, and MongoDB access.
bcrypt	Hash passwords before storage; never store plain-text passwords.
JWT or secure session	Authentication mechanism with expiry and secure handling, chosen consistently during implementation.
Git & GitHub	Version control, code history, and backup.
Postman	Test backend REST API endpoints.
MongoDB Compass	Inspect MongoDB collections and documents.
Optional integrations	AI API, maps, email/SMS, image storage, payment sandbox, and 360° media. Availability depends on credentials, costs, approvals, and time.

3.What Problems My Project Solves

EasyProperty aims to bring important real-estate activities into one connected platform. Although existing platforms provide property search and listings, users may need to handle comparisons, enquiries, viewing requests, and follow-ups separately. EasyProperty will focus on connecting these activities through the following features.

1. Role-Based Dashboards: Separate dashboards for buyers, sellers, agents, landlords, investors, and administrators will provide the tools each user needs.
2. Property Comparison: Users can compare properties by price, area, location, and property type to make better-informed decisions.
3. Enquiry and Viewing Tracking: Buyers and tenants can submit enquiries and viewing requests and track their status without depending entirely on manual follow-ups.
4. Category-Specific Listing Forms: Dedicated forms for houses, plots, rental properties, commercial spaces, and agricultural land will collect relevant property details.
5. Landlord Management: Landlords can manage rental listings and basic enquiry records through their dashboard.
6. AI Property Guidance: An AI assistant will explain common property terms and due-diligence steps, helping users understand important considerations before making decisions.
7. Listing Risk Warnings: The system will flag incomplete or unusual listing information for further review. These warnings will not prove fraud or verify legal ownership.

8. Admin Moderation: Administrators can review listings, approve or reject submissions, request corrections, and handle reports through a centralized dashboard.

9. Investor Comparison and 360° Tours: Investors can compare property details, while 360-degree tours can help users explore supported properties visually before arranging a visit.

Existing Platforms and Proposed Focus

Platforms such as Zameen.com and other property portals already provide property browsing, sale/rent listings, search filters, and agent connections. EasyProperty will not claim those common features are unique. Its proposed focus is the combination of role-specific dashboards, connected enquiries and viewing requests, simple investor comparison, landlord tools, category-specific listing forms, admin moderation, and informational AI-assisted guidance. This is a proposed value proposition, not a claim that EasyProperty is already better or has more users.

4.Main Project Modules

No.	Module	Main functions
1	Public Website & Property Discovery	Home page, navigation, category pages, search results, filters, property details, and responsive design.
2	Accounts & Role-Based Access	Registration, login, password hashing, sessions/tokens, protected routes, and role permissions.
3	Property Listing Management	Category-specific fields, create/edit/delete listing, photos, ownership checks, and listing status.
4	Buyer/Tenant Tools	Favourites, compare, enquiries, viewing requests, and request-status tracking.
5	Seller & Agent Dashboard	Manage own listings, review status, incoming enquiries, and listing updates.
6	Landlord Rental Management	Rental listings, basic tenant/enquiry records, and rental status tracking.
7	Investor Comparison	Compare asking price, location, area, type, and clearly labelled estimated figures.
8	Admin & Moderation	Review listings, approve/reject/request changes, process reports, and monitor status.
9	AI Guidance & Risk Warnings	General property-term explanations, due-diligence checklist, and rule-based/AI-assisted warning indicators for human review.
10	360° Tours, Notifications & Payment Demo	Compatible 360° media viewing, in-app notifications, and mock/sandbox payment demonstration.
11	Database & API Layer	MongoDB models, Express REST APIs, validation, error handling, and data access.
12	Testing, Deployment & Documentation	Functional/security checks, bug fixes, setup guide, user guide, project report, and final presentation.

5. Weekly Module Breakdown (5 Months)

Week	Module / Work	Detailed scope and deliverable	Deadline
1	Planning, Requirements & Project Setup	Confirm project scope, user roles, required pages, priority features, database choice, and acceptance criteria. Create Git repository and project folders. Deliverable: Approved scope, feature list, folder structure.	16 Oct 2026
2	Frontend: Home Page & Shared Layout	Build responsive header, navigation, footer, hero section, property-category links, featured listings, search entry point, and reusable buttons/cards. Deliverable: Home page and shared responsive layout.	23 Oct 2026
3	Frontend: Login, Registration & Role Screens	Create login, sign-up, forgot-password UI and initial dashboard shells for buyer/tenant, seller/agent, landlord, investor, and admin. Use sample data; no real authentication yet. Deliverable: Account pages and role dashboard layouts.	30 Oct 2026
4	Frontend: Property Browse & Category Pages	Create browse page and category pages for homes, plots, rental properties, commercial spaces, agricultural land, and investment properties. Add sample property cards. Deliverable: Linked category and browse pages.	06 Nov 2026
5	Frontend: Search, Filters & Property Details	Build search results, location/keyword search UI, property type, sale/rent, price, area and sorting filters; property details with images, features, location text, and contact CTA. Deliverable: Search results and details pages.	13 Nov 2026
6	Frontend: Favourites, Compare & Enquiries	Build Favourites/save UI, compare-properties page, enquiry form, viewing-request form, success/error states, and validation. Investor comparison uses visible listing facts and user-supplied estimates. Deliverable: Buyer/tenant and investor frontend flows.	20 Nov 2026
7	Frontend: Seller, Agent & Landlord Pages	Build create/edit listing pages, my listings table, listing status, enquiries inbox, landlord rental records, and tenant/enquiry tracking screens. Deliverable: Seller/agent/landlord page set.	27 Nov 2026
8	Frontend: Admin, Trust & Advanced UI	Build admin dashboard, listing review queue, approve/reject/request-changes controls, reports, document-status UI, AI guidance/risk-warning panel, 360° tour placeholder/viewer shell, payment-demo screen. Link and test all frontend pages. Deliverable: Complete clickable responsive frontend prototype.	04 Dec 2026
9	React Structure & Frontend Integration	Organize UI into React + Vite components, add routes, shared layouts, state patterns, form validation, responsive checks, and configuration. If already used from Week 2, refactor and stabilize. Deliverable: Structured React application.	11 Dec 2026
10	Backend Foundation: Node.js & Express	Set up Node.js project, Express server, REST API routes, middleware, environment variables, centralized error handling, health-check endpoint, and API documentation outline. Deliverable: Running backend API skeleton.	18 Dec 2026

11	MongoDB Connection & Data Models	Set up MongoDB local/cloud development database and Mongoose. Define User, Property, Favourites, Enquiry, Viewing Request, Report, and moderation/status models; add sample seed data and indexes. Deliverable: Connected MongoDB with tested schemas.	25 Dec 2026
12	Authentication Backend & Authorization	Implement registration/login APIs, bcrypt password hashing, validation, secure token or session handling, logout/expiry behavior, protected routes, and server-side role checks. Connect login/register UI. Deliverable: Working authentication and role permissions.	01 Jan 2027
13	Property CRUD APIs & Image Handling	Create listing APIs to add, view, edit, and delete properties; validate required fields; enforce listing ownership; store image references or use a suitable upload service if available. Deliverable: Connected property management.	08 Jan 2027
14	Search/Filter APIs & Property Pages	Implement database-backed search, category, location, sale/rent, price, area, and sorting filters. Connect browse, search, and detail pages to API and handle loading/empty/error states. Deliverable: Working database-backed property search.	15 Jan 2027
15	Favourites, Enquiries & Viewing APIs	Implement save/remove favorites, enquiry creation and status, viewing requests, user-specific records, and connect buyer/tenant pages to backend. Deliverable: Connected buyer/tenant workflows.	22 Jan 2027
16	Seller/Agent & Landlord Workflows	Connect seller/agent dashboards to owned listings, listing status, enquiries and edits. Add basic landlord rental record functions and access controls. Deliverable: Connected seller, agent and landlord dashboards.	29 Jan 2027
17	Admin Moderation & Reports	Build admin-only APIs and pages for listing review, approve/reject/request changes, reports, document-status tracking, and basic dashboard counts/status history. Deliverable: Working moderation and report workflow.	05 Feb 2027
18	Investor Comparison & AI Guidance	Connect investor comparison to saved/listing data and label calculations as estimates. Add AI API integration if access is available; otherwise implement a clearly labelled informational FAQ/checklist and rule-based warning demo. Deliverable: Investor tools and guidance/risk-warning prototype.	12 Feb 2027
19	360° Tours, Notifications & Payment Demo	Add compatible 360° media viewer, in-app status notifications, and payment sandbox/mock workflow. Do not process or store real card data. If provider access is unavailable, document a clearly labelled demo. Deliverable: Advanced feature demonstrations.	19 Feb 2027
20	Integration, Security & Functional Testing	Test core flows end-to-end: registration, roles, listing ownership, search, favourites, enquiries, viewings, moderation, validation, error states, and responsive UI. Fix high-priority defects. Deliverable: Test results and corrected release candidate.	26 Feb 2027
21	Deployment, Documentation & Final QA	Prepare setup instructions, environment variable example, database setup/seed steps, API notes, user guide, final screenshots, deployment configuration, regression test, and backup copy. Deliverable: Deployable candidate and complete documentation.	05 Mar 2027
22	Final Demonstration & Submission	Run final demo, verify required acceptance criteria, finish report/presentation, capture evidence, and submit source code and documentation by the project end date. Deliverable: Final completed project and presentation.	09 ar 2027

6.Scope Limits and Risk Management

Core functionality takes priority over paid/external integrations. AI API access, maps, email/SMS, cloud image storage, live payment processing, and official property-record access may require credentials, fees, or approval. If an integration is unavailable, EasyProperty will include a clearly labelled mock/demo or documented integration point. AI outputs and uploaded documents are not legal proof; risk flags do not guarantee fraud detection. The project will not store real payment-card details or promise investment returns.

7.Conclusion

EasyProperty is planned from 9 October 2026 to 9 March 2027. The schedule moves from page-by-page frontend work to backend APIs, MongoDB integration, authentication, connected user workflows, advanced demonstrations, testing, and final handover. The objective is to complete and demonstrate all core modules within five months, with external services represented transparently if access is unavailable.